29. Phenol dimerises in benzene having van't Hoff factor 0.54. What is the degree of association ?

a) 0.46 b) 92

c) 46 d) 0.92

30. Assertion : An ideal solution obeys Raoults Law

Reason : In an ideal solution, solvent-solvent as well as solute-solute interactions are similar to solute-solvent interactions.

a) both assertion and reason are true and reason is the correct explanation of assertion

b) both assertion and reason are true but reason is not the correct explanation of assertion

c) assertion is true but reason is false

d) both assertion and reason are false

II. Write brief answer to the following questions.

31. Define (i) molality (ii) Normality

32. What is a vapour pressure of liquid?

What is relative lowering of vapour pressure?

33. State and explain Henry's law

34. State Raoult law and obtain expression for lowering of vapour pressure when nonvolatile solute is dissolved in solvent.

35. What is molal depression constant? Does it depend on nature of the solute ?

36. What is osmosis?

37. Define the term 'isotonic solution'.

38. You are provided with a solid 'A' and three solutions of A dissolved in water - one saturated, one unsaturated, and one super saturated. How would you determine which solution is which ?

39. Explain the effect of pressure on the solubility.

40. A sample of 12 M Concentrated hydrochloric acid has a density 1.2 gL^{-1} Calculate the molality

41. A 0.25 M glucose solution at 370.28 K has approximately the pressure as blood does what is the osmotic pressure of blood ?

42. Calculate the molality of a solution containing 7.5 g of glycine (NH_2-CH_2-COOH) dissolved in 500 g of water.

43. Which solution has the lower freening point?10 g of methanol (CH_3OH) in 100g of water (or) 20 g of ethanol (C_2H_5OH) in 200 g of water.

44. How many moles of solute particles are present in one litre of 10^{-4} M potassium sulphate?

45. Henry's law constant for solubility of methane in benzene is $4.2x10^{-5}$ mm Hg at a particular constant temperature At this temperature. Calculate the solubility of methane at i) 750 mm Hg ii) 840 mm Hg

46. The observed depression in freezing point of water for a particular solution is 0.093^0C. Calculate the concentration of the solution in molality. Given that molal depression constant for water is 1.86 KKg mol^{-1}

47. The vapour pressure of pure benzene (C_6H_6) at a given temperature is 640 mm Hg. 2.2 g of non-volatile solute is added to 40 g of benzene. The vapour pressure of the solution is 600 mm Hg. Calculate the molar mass of the solute?

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